

FEATKT: FEATURE-DRIVEN GRADIENT BOOSTING FOR MULTI-STEP KNOWLEDGE TRACING

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INTRODUCTION

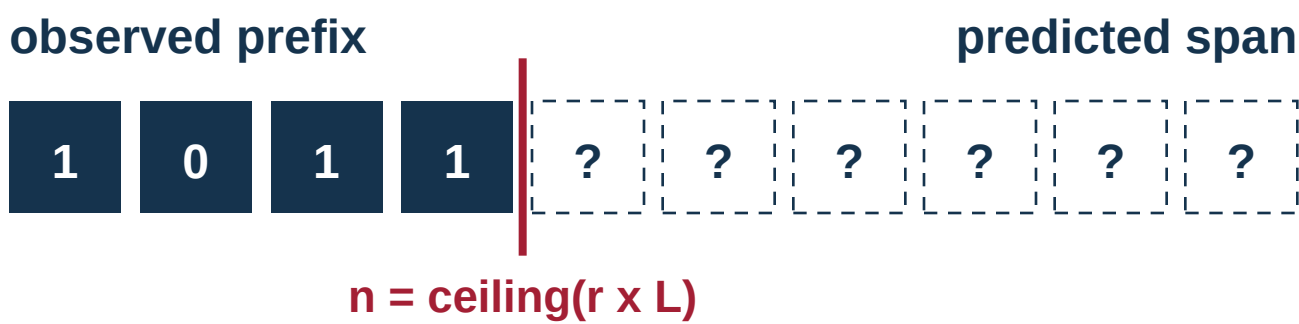
Knowledge tracing (KT) predicts whether a student will answer a question correctly from their learning history.

Most KT models are evaluated one step ahead: the previous answer is revealed before the next prediction. Real deployments often require predicting an entire future sequence without new feedback - for example, weekly syllabus planning or early identification of at-risk students.

pyKT calls this **non-accumulative multi-step (NAMS)** prediction [5]. Once prediction starts, a sequence model receives no new responses, making the information in the input features especially important.

EVALUATION PROTOCOL

For a sequence of length L and prefix ratio r , the model observes the first $\text{ceiling}(r \times L)$ interactions and must predict all remaining outcomes without receiving additional correctness labels.



Data: 5,000 EdNet-KT1 students [6], split by user into 4,000 training and 1,000 testing students. No student's interactions appeared in both sets.

Metric: AUC averaged across prefix ratios 0.2, 0.4, 0.6, 0.8, and 0.9.

PROPOSED MODEL: FEATKT

FeatKT is a LightGBM gradient-boosted decision tree model [7]. It predicts each interaction using five leakage-checked features:

- Concept identifier:** the concept or skill tested by the question.
- Question difficulty:** the standardized training-set error rate for the question.
- Log elapsed time:** $\log(1 + \text{time spent answering the previous question})$.
- Log lag time:** $\log(1 + \text{time between consecutive interactions})$.
- Frozen prefix accuracy:** the student's observed-prefix accuracy, calculated once at the prediction boundary and never updated using future responses.

Why freeze accuracy?

Updating this feature during prediction would require answers from the future span and violate the NAMS protocol. The frozen value preserves a student-level summary without leakage.

SELECTED REFERENCES

[1] A. T. Corbett and J. R. Anderson, 'Knowledge Tracing: Modeling the Acquisition of Procedural Knowledge,' *User Modeling and User-Adapted Interaction*, 1994.

[2] C. Piech et al., 'Deep Knowledge Tracing,' *NeurIPS*, 2015.

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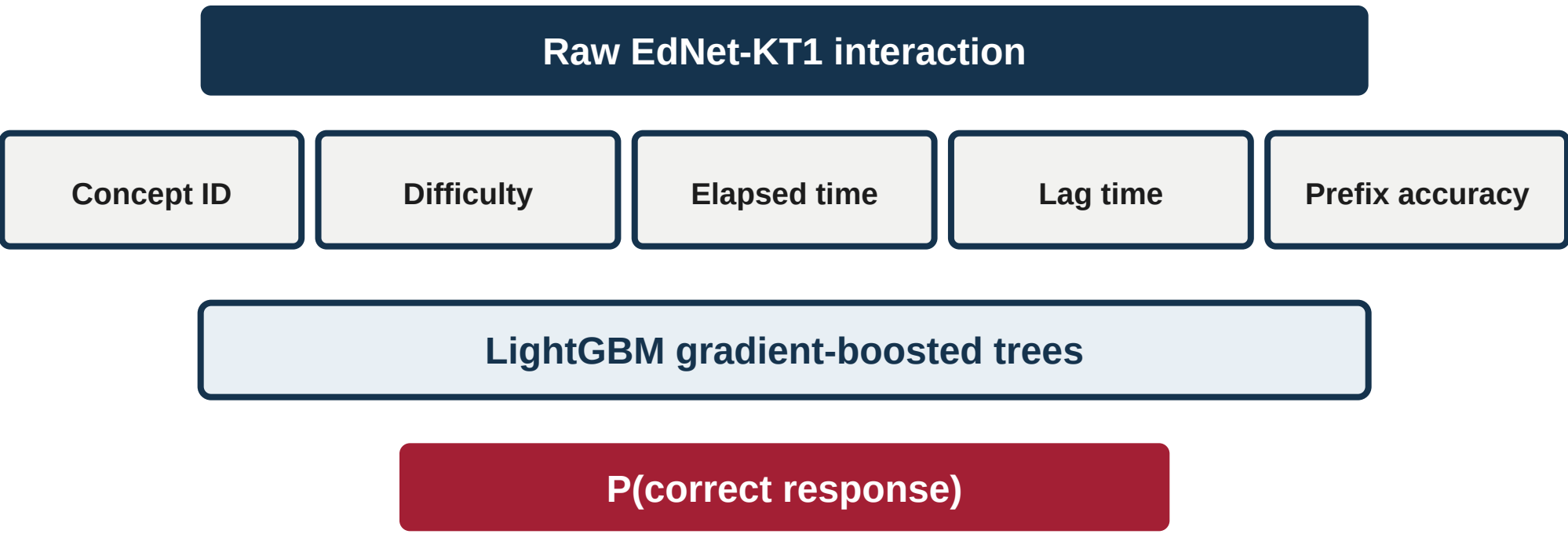
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[5] Z. Liu et al., 'pyKT: A Python Library to Benchmark Deep Learning Based Knowledge Tracing Models,' *NeurIPS*, 2022.

[6] Y. Choi et al., 'EdNet: A Large-Scale Hierarchical Dataset in Education,' *AIED*, 2020.

[7] G. Ke et al., 'LightGBM: A Highly Efficient Gradient Boosting Decision Tree,' *NeurIPS*, 2017.

ARCHITECTURE



Each interaction is converted to the five features, scored by boosted trees, and mapped to a probability of a correct response.

RESULTS

Model	Type	Avg. AUC
DKT	Recurrent	0.594
sparseKT	Sparse attention	0.646
simpleKT	Transformer	0.666
AKT	Difficulty-aware	0.666
Transformer + features	Transformer	0.726
FeatKT	Boosted trees	0.747

FeatKT achieved the highest average AUC (0.747) and led at every prefix ratio. Adding the engineered features to a transformer raised average AUC from 0.666 to 0.726, showing that the feature set itself drove most of the gain.

MODEL VALIDATION

A label-shuffled control achieved 0.5093 AUC, consistent with chance and providing no evidence of target leakage. A paired bootstrap over test users found a positive FeatKT advantage at every prefix ratio ($p < 0.001$).

Prefix	Gap	95% CI	Favoring FeatKT
0.2	0.024	[0.014, 0.034]	100.0%
0.4	0.027	[0.018, 0.037]	100.0%
0.6	0.019	[0.014, 0.025]	100.0%
0.8	0.022	[0.014, 0.029]	100.0%
0.9	0.012	[0.006, 0.020]	99.9%

Paired-bootstrap AUC gap versus the feature-matched transformer; 1,000 resamples over test users.

CONCLUSION

Under NAMS prediction, input features had a greater impact on performance than model architecture. FeatKT's five features supported stronger long-horizon predictions with a simpler model.

Limitations: 5,000 EdNet users and one train-test split. Future work should use full EdNet, ASSISTments, additional leakage-checked features, and per-feature ablations.

0.747 AUC

Highest average across all tested models

Code: github.com/sahishnu/FeatKT